# NutriCounter - Smart Food Assistant

**A Web and Mobile Application for Intuitive and Accurate Nutritional Tracking**

[redacted]

ITS 120L

July 7, 2025

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# Introduction

## 1.1 Background of the Study

In recent years, especially during the pandemic that we've all experienced, public awareness regarding health and nutrition has been mostly neglected. A lot of people weren't able to go outside and do their daily activities that require physical movement and there are also a lot of people that aren't into gyms or do not exercise, and thus a lot of people became either overweight or are obese as a result. With the rise of the internet and access to endless amounts of information, most have the capabilities in searching and understanding the impact of their diet on their overall well-being but are lazy or unwilling to do so. We now live in an age where most people are conscious of their physical well-being, especially with the rise of social media applications, where tracking nutrients, managing caloric intake, and making/formulating diets with fitness goals are usually done and are encouraged.

However, this availability of information on the internet regarding diets and food calories and so on highlighted a significant problem in the abundance of tools that a person can access. The convenience of a smart AI assistant that will aid you in tracking your nutrition and food intake comes into major play because of the issue mentioned above. This abundance of information online usually stumps most people so finding the "accurate" nutritional information isn't just about looking up an ingredient on the internet. It will account for portion sizes that reflect real-life cooking (Such as "a cup of chopped carrots," not "128 grams"). This is what our project, NutriCounter, aimed to solve. It is a Web and Mobile application which is catered to the user's real-world eating habits.

## Statement of Objectives

### 1.2.1 General Problem

The problem that our project aims to solve is the difficulty and inaccuracy of tracking daily nutritional intake. Current nutrition calculators, that are available on the internet, primarily only rely on food weight (in grams or ounces etc…), which is impractical or difficult for everyday users who do not weigh every component of their meals or don't have the ability to do so. This leads to inaccurate or wrong estimates, and will result in them eventually abandoning their food tracking because of those inconsistencies. With the help of AI, this project will try to aid the users in providing the best estimate of how much nutrients they are taking exactly.

With the numerous amounts of home-cooked meals and slight variations in food preparation that most households typically do, most aren't able to accurately weigh the food they are taking and fail to capture the nutritional content of what they are consuming.

### 1.2.2 Specific Problem

Users struggle to translate their real-world meals into the very rigid input formats required by existing apps available on the internet, and most are pay-walled. It's the lack of intelligent and flexible input processing that makes users either guess or give up entirely. There is a lack of a user-friendly application that can intelligently interpret descriptive portion sizes (e.g., "1 medium tomato," "a handful of almonds") and automatically adjust nutritional values based on specified preparation methods (e.g., raw, steamed, sautéed in oil). Users can also add the exact weight and portion if they can.

### 1.2.3 General Objective

The general objective of our project is to develop a web and mobile application, "NutriCounter," that leverages Artificial Intelligence and data online to automate and streamline the process of nutritional calculation, no matter the food they have. This will make tracking more intuitive, accurate, and sustainable for users, which helps users make informed food choices and align their meals with their health goals or dietary restrictions.

### 1.2.4 Specific Objectives

The specific objectives are:

- To design and develop a user-friendly web and mobile interface for easy meal logging and nutritional review.
- To integrate with generative APIs like Gemini and which will help users in calculating their daily nutritional intake and give verified data for a wide variety of foods.
- To implement AI system prompts for:
    - Interpreting user inputs like "1 cup chopped carrots" or "a drizzle of olive oil" into standardized measurements.
    - Identifying food items, quantities, and preparation methods from a single text entry and formats it into an answer that the website can accept and interpret.

## 1.3 Scope and Limitation

**Scope:**

- Focuses on all ranges of food (fruits, vegetables, meats, grains, you name it).
- Analyzes macronutrients (protein, carbohydrates, fats, fiber) and key micronutrients (such as, sodium, sugar, select vitamins).
- Provides a web-based interface that works on PC and mobile devices (iOS/Android).
- Allows users to log meals using descriptive text inputs.

- Provides personalized reports based on user-defined health goals (e.g., calorie targets, etc…)
- Uses Gemini's API for calculation and analysis based on the aforementioned necessities above.

**Limitation:**

- The accuracy depends on the quality and input of data from the users and other nutritional databases available online.
- Estimates for ambiguous, non-standardized portions (e.g., "a handful," "a splash") will be based on standardized averages and are inherently approximations.
- The application is an informational tool and should not be treated as a substitute for professional medical or dietetic advice.

## The Proposed System

### 2.1 System Overview

Since NutriCounter is a web and mobile application, we aimed to provide a user-friendly interface for users to interact with. Below is the outline of the system:

- User Input: The user enters a description of their meal or food item into the input field (e.g., "2 scrambled eggs with a slice of whole wheat toast and half an avocado").

- Input Processing: NutriCounter uses Google's Gemini API to parse the user's text. It identifies the individual food items ("eggs," "whole wheat toast," "avocado"), the quantities ("2", "a slice," "half"), and preparation methods ("scrambled").

- AI-Powered Analysis:
  - Data Retrieval: The system (Gemini API) queries its integrated search function for the standard nutritional profile of each identified food item.
  - Quantity Calculation: It converts the descriptive quantities into standard units (e.g., "2 eggs" -> 2 large eggs; "a slice of toast" -> ~28g) unless the user adds more details about the exact portion of the wood and how much it weighs.
  - Preparation Adjustment: The API will format the nutritional values based on the preparation method for the website to interpret and understand the results given by the API. For "scrambled eggs," it may add a default amount of fat and calories for oil/butter, whereas steamed vegetables (like carrots etc…) would have its vitamin C content slightly reduced.
  - Personal AI Health Coach: The API will take into account the user's BMI to formulate diet and workout plans that would suit his/her body. It would also counsel you on whether to lose, gain or maintain your current BMI.

- Nutritional Aggregation and Logging:
  - It then combines the calculated nutritional data from all food components to provide a total for the meal (total calories, protein, carbs, fat).

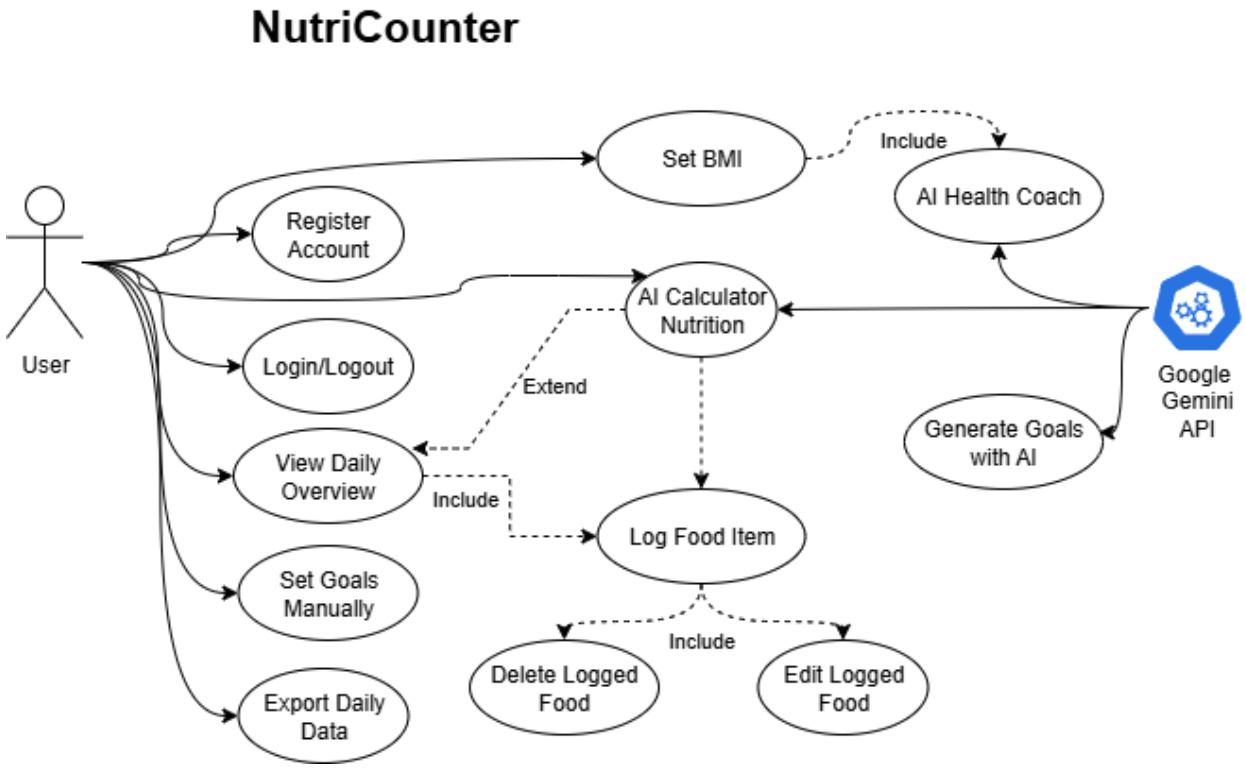
- Data Presentation:
  - The nutritional breakdown of the meal is presented to the user in a clear, organized, and user-friendly format. The user can view the meal total as well as the breakdown for each individual nutritional value.
- Personalization:
  - Users can set personal health goals (e.g., daily calorie limit, protein target, carb limit). The system logs the meal to their daily diary and updates their progress

towards these goals, providing simple feedback and visualizations.

The complete source code / repository for this project is publicly available on GitHub and can be accessed for review.

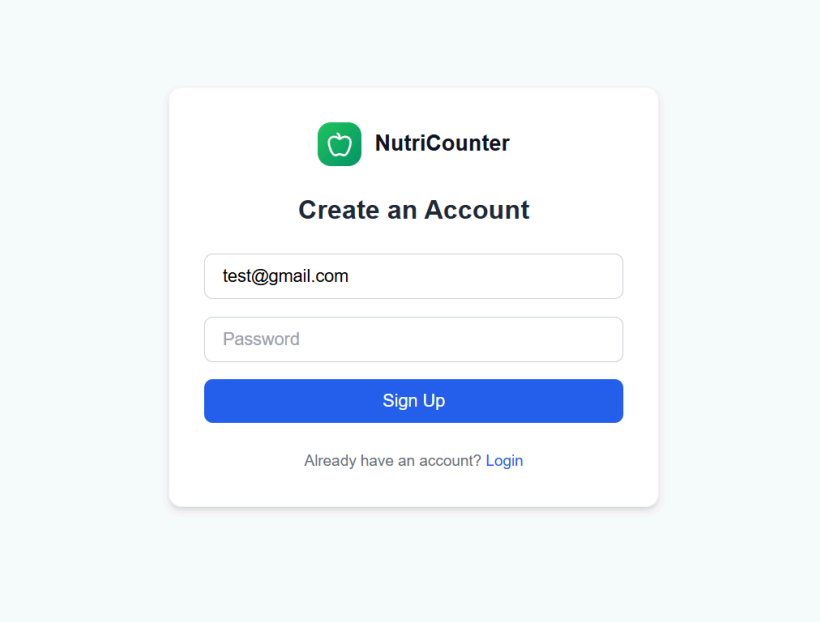
**Source Code:** https://github.com/martyUAE/NutriCount#

### 2.2 Use Case Diagram

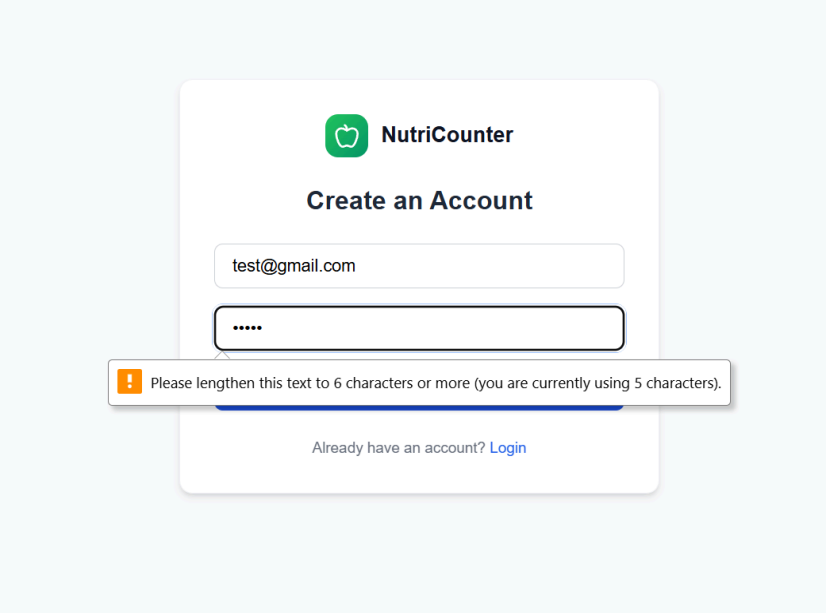



### 2.3 Screen Layout

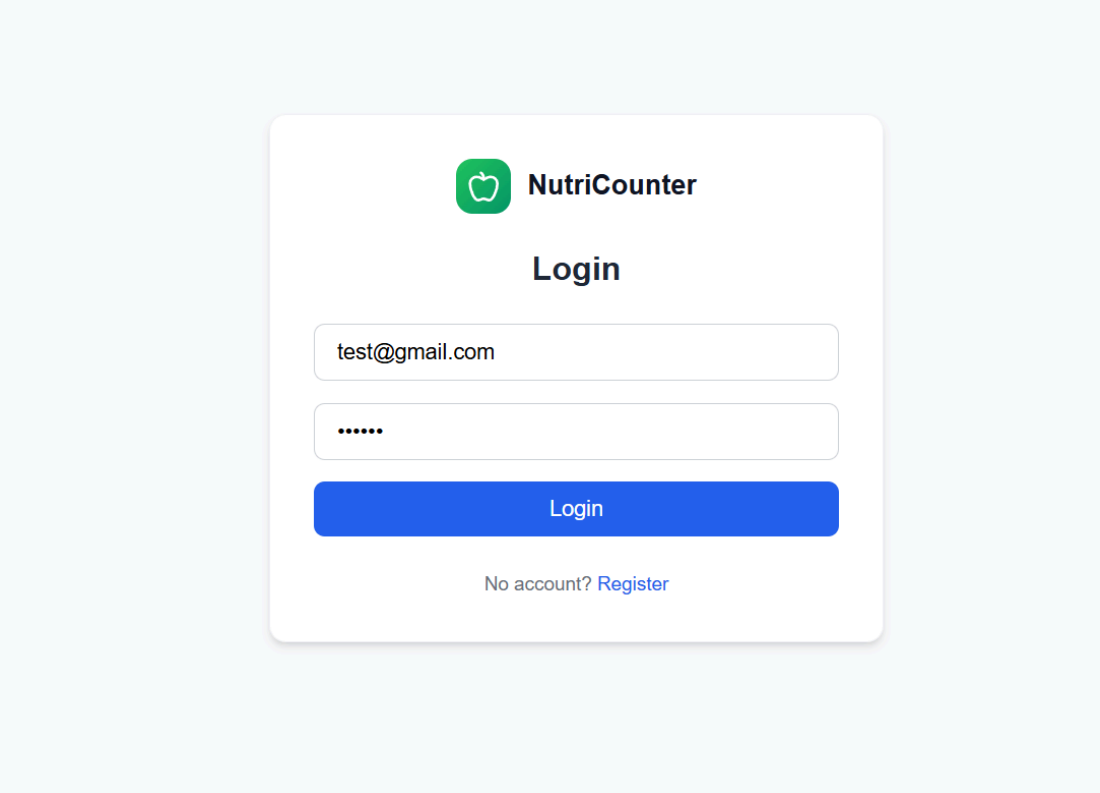
1. How to create an account for the User:
    a. Users must input an email address.



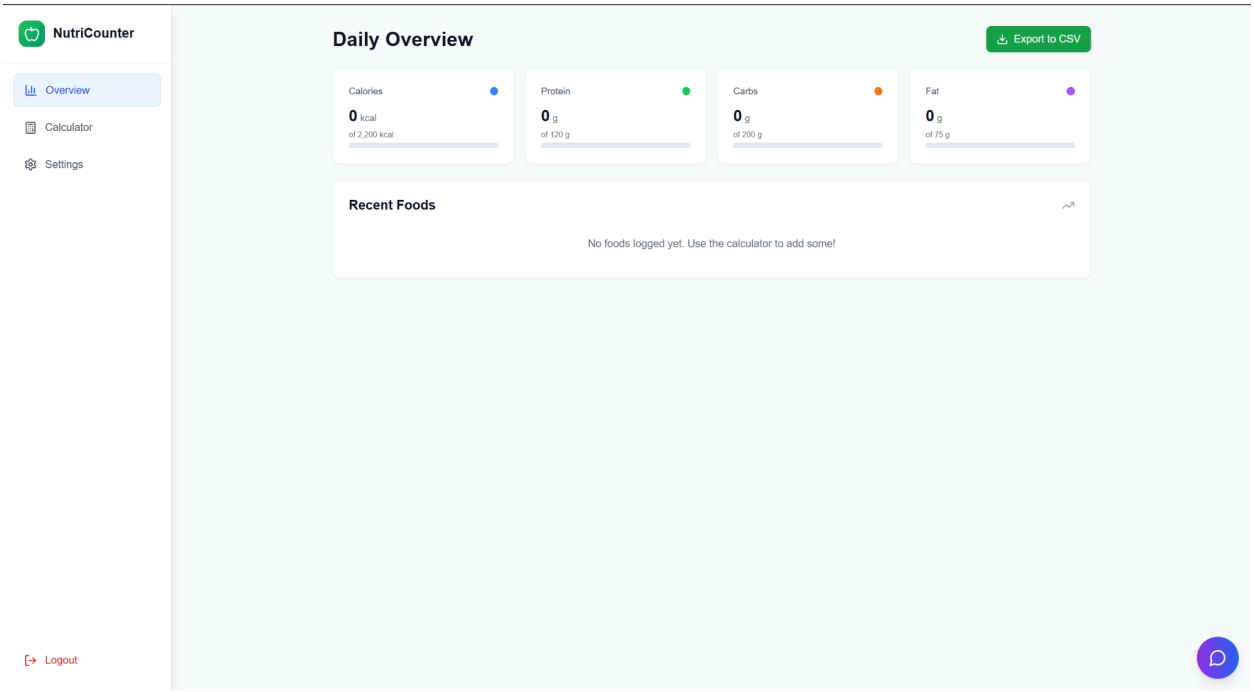

b. Users need to have a minimum of 6 characters for their password to get accepted by the system.




2. How to login on a created account for the User:
    a. Once the User had successfully created an account, they should be automatically logged in. If in the event that they aren't, they will be redirected to the login page where they would insert their account credentials.

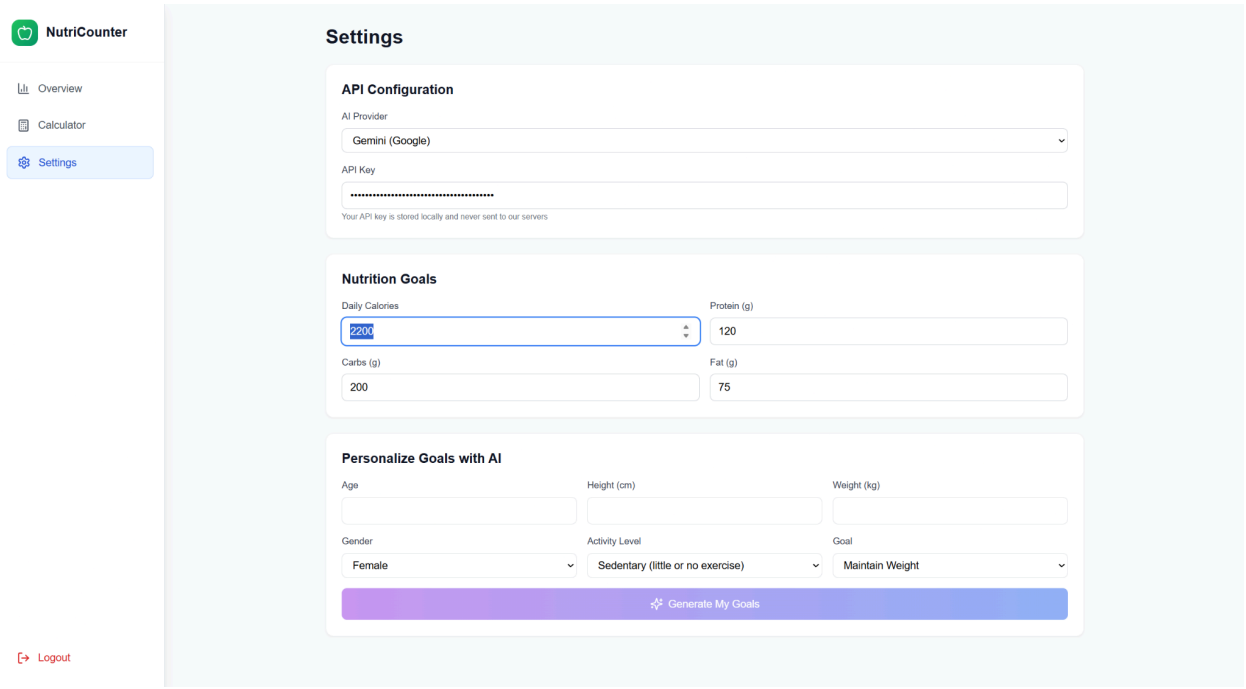



b. Users will be redirected to the Overview page once they are successfully logged in.

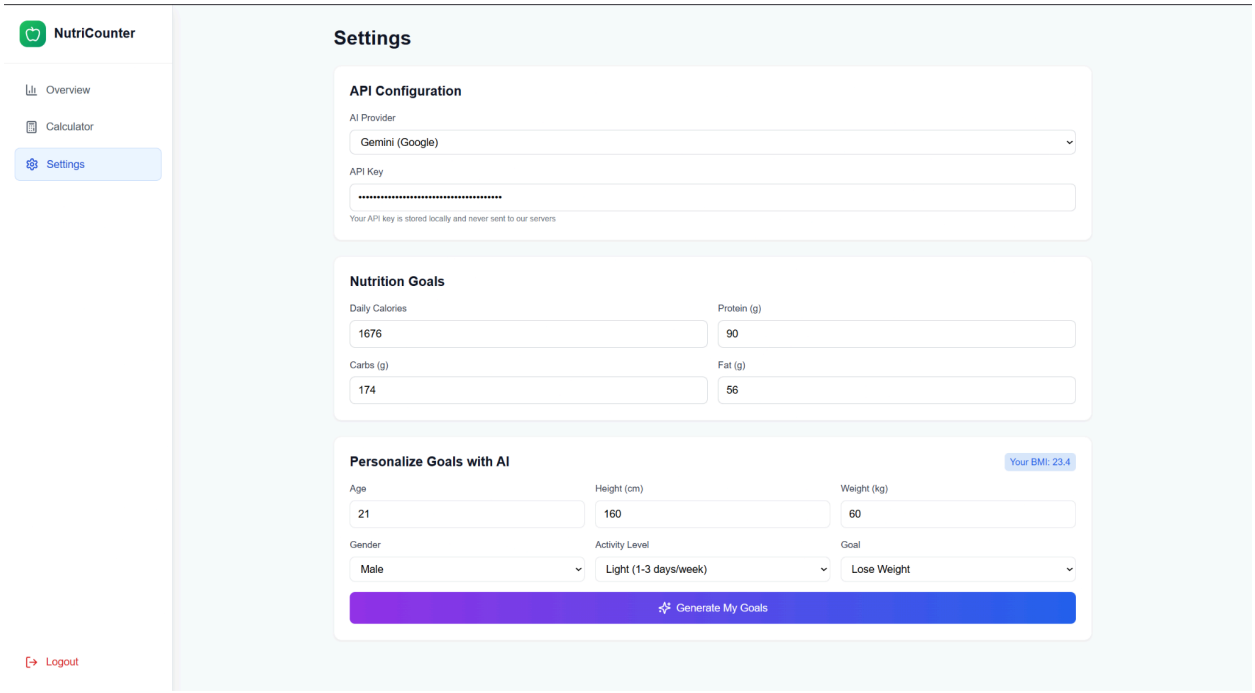



3. What the Overview is about:

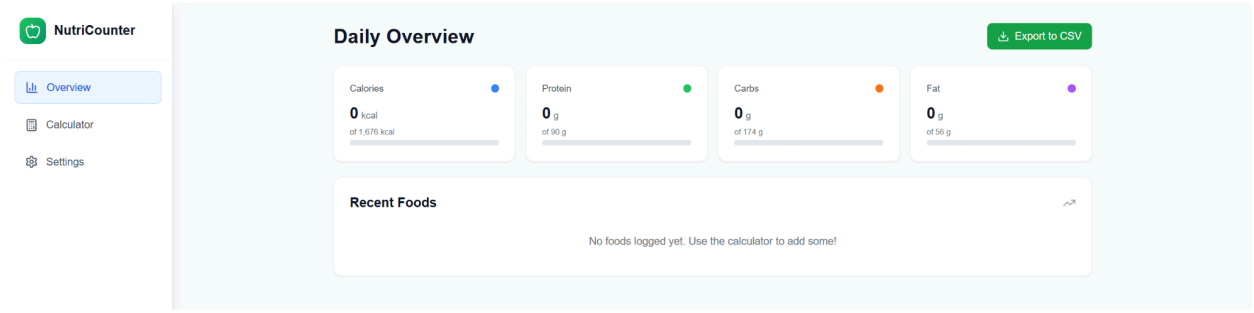
a. Once the user is in the Overview page, they will see the Daily Overview, Export to CSV, Recent Foods and Chat Bot. The" Daily Overview" section is about the User's daily nutritional goals. Users can set their own nutritional goals if they go to the Settings page.




b. Users can also let AI generate their personal nutritional goals by inserting their Age, Height and Weight which is calculated to get their BMI and then sent to the API to calculate the recommended nutritional goals they need based on their calculated BMI.

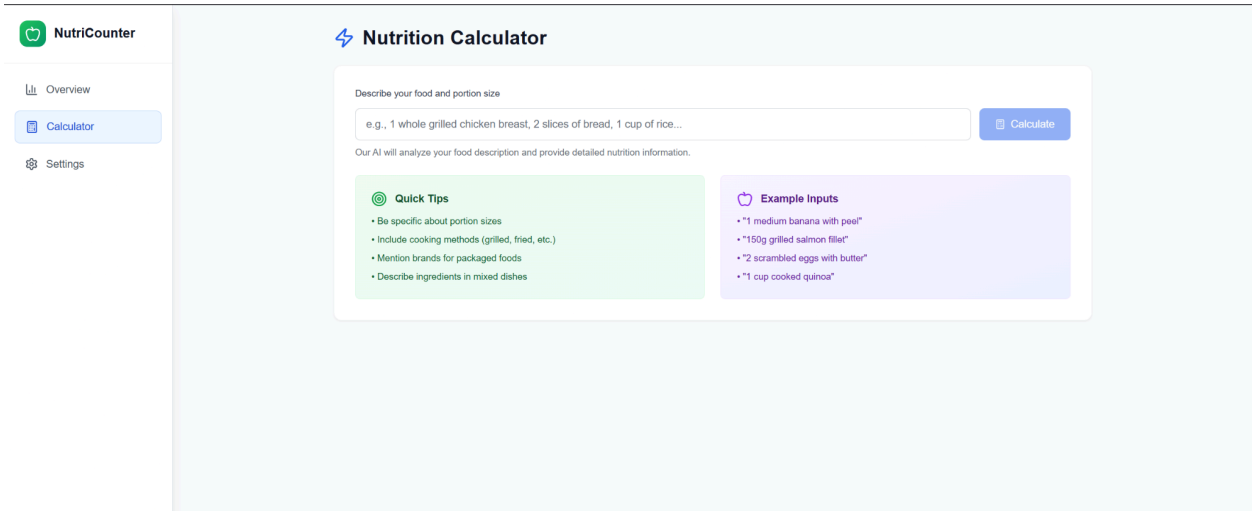



c. After clicking “Generate My Goals”, the changes would be reflected on the Nutrition Goals as well as on the Overview page.

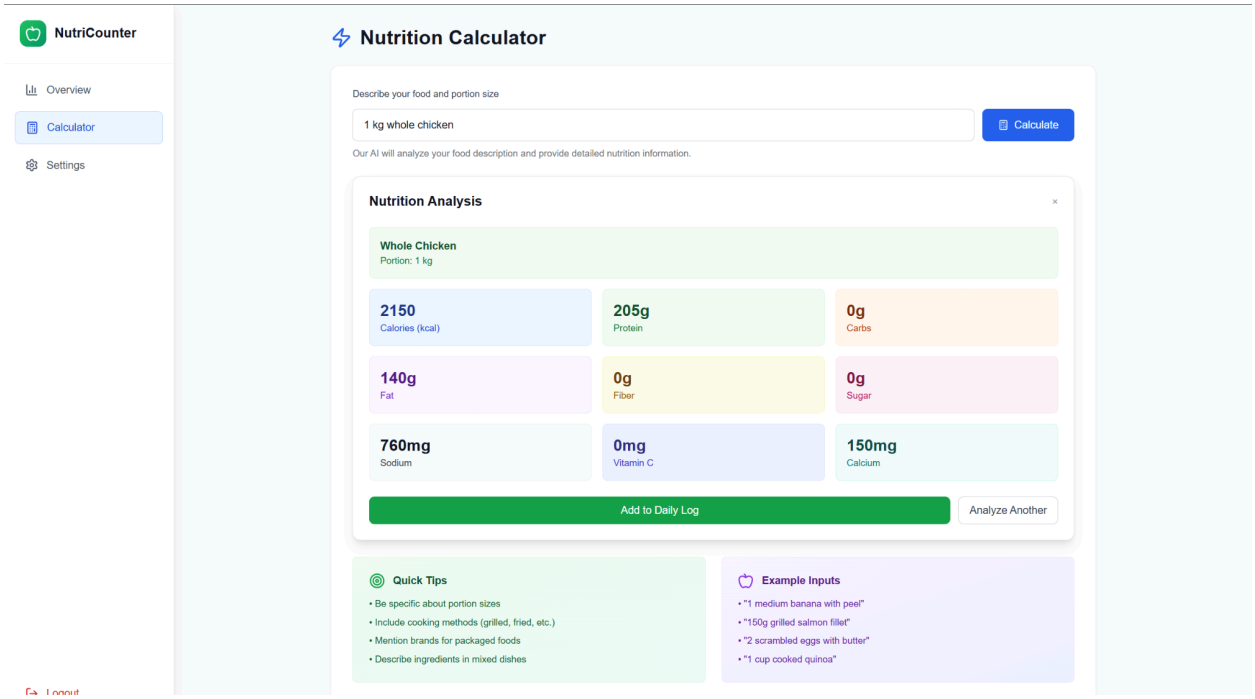



4. How to use the Nutrition Calculator:

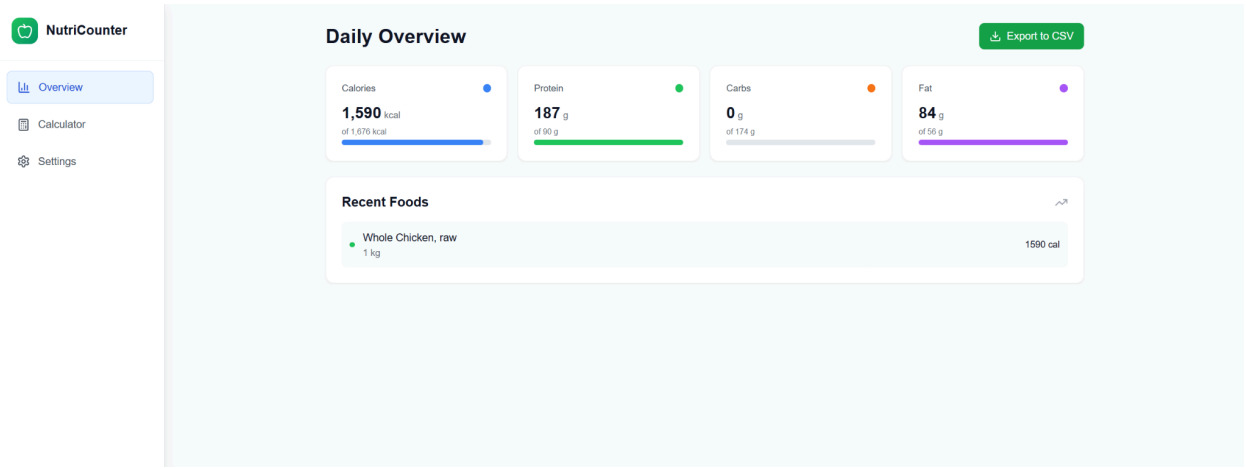
a. When the user clicks on the “Calculator” page, it would bring up the search function which the users can use to find their specific food.



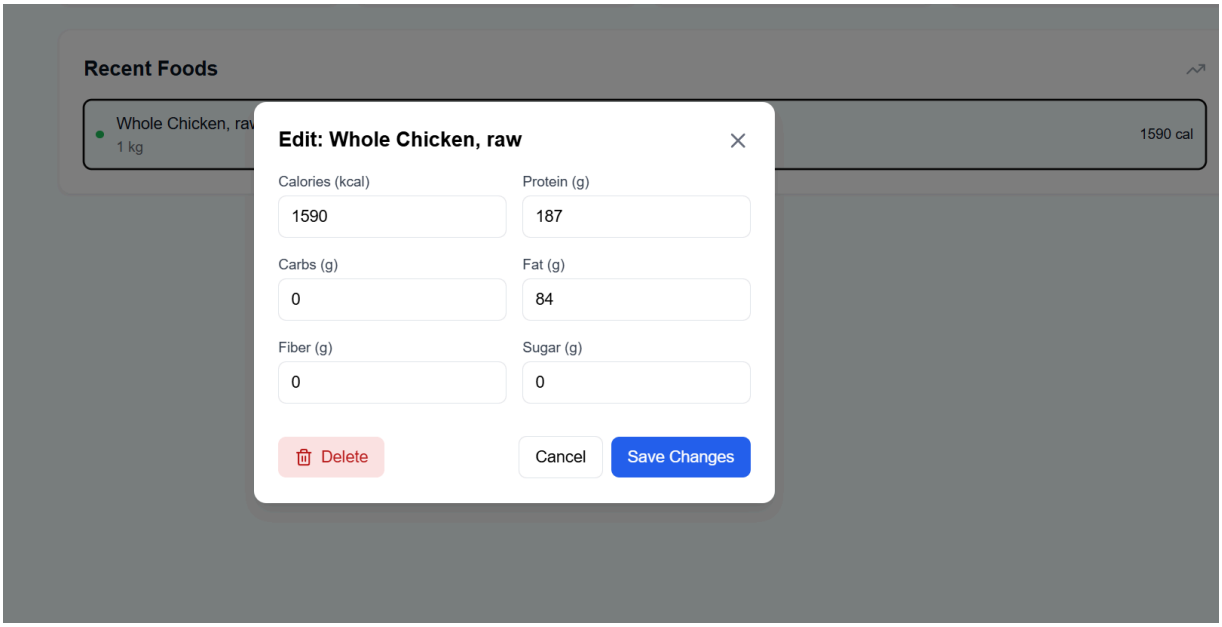

b. In this Nutritional Calculator, you can search for specific food names e.g. 4 cups of Rice, 1 kg whole chicken roasted, etc… For more accurate results, it is best to be specific on the measurement but a rough estimate could also work.



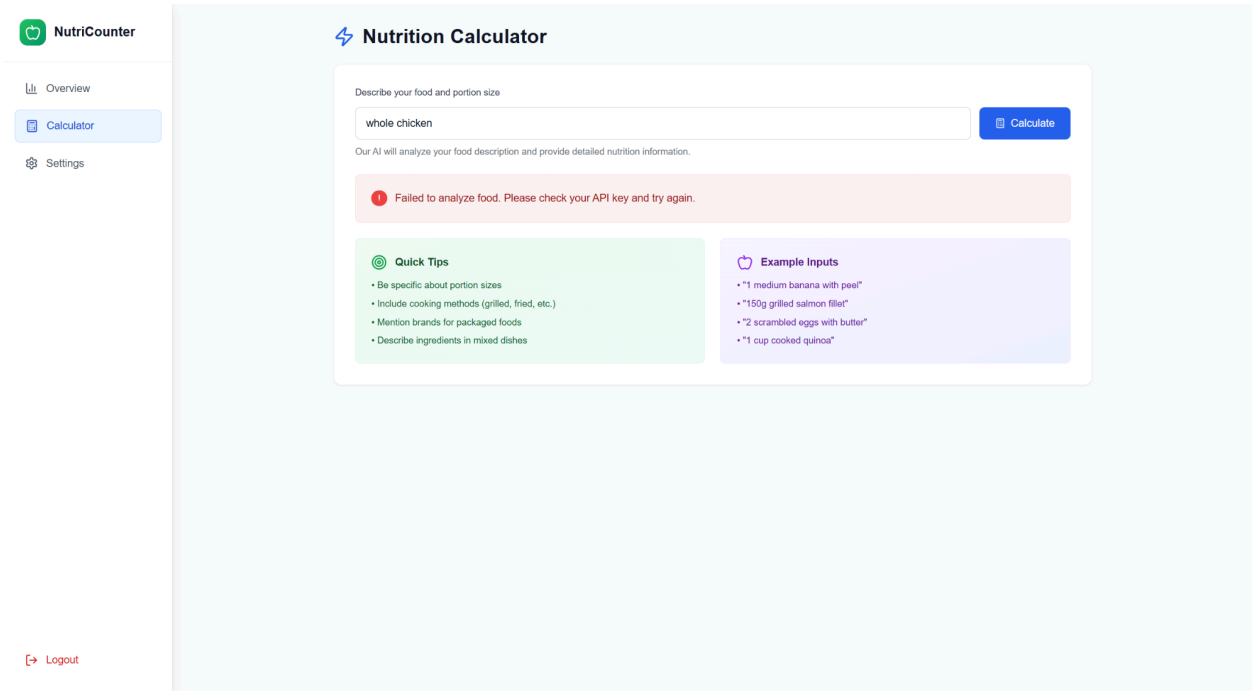

i. If the User clicks on Add to Daily Log button, it would appear on the “Overview” page and their Daily Overview section would also get updated based on the calculated nutrients




ii. The user is able to edit the already added items on the “Recent Foods” and they can also delete it from the table. The user only needs to click the specific food item and a popup menu would appear.

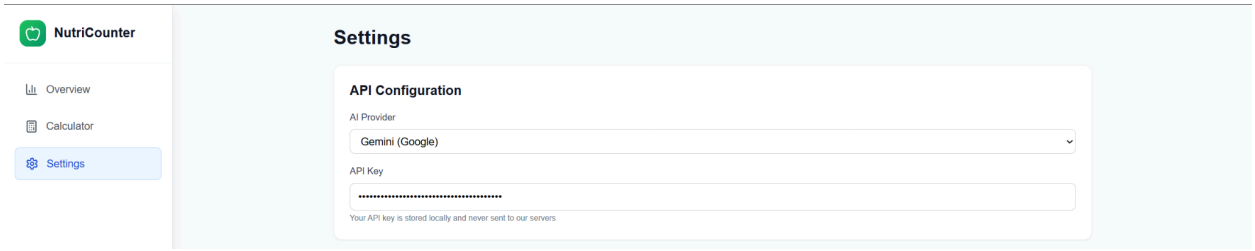



c. If the food is not clearly defined, it would return an error and would prompt the user to search again.

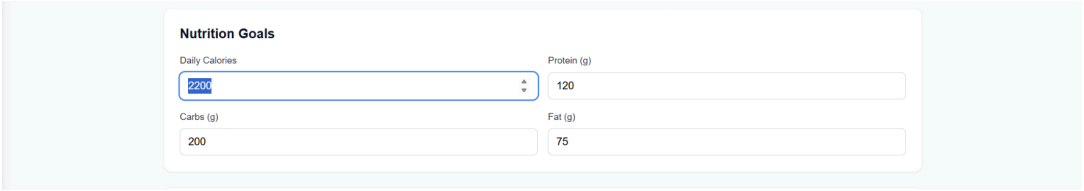



5. What the settings are about:

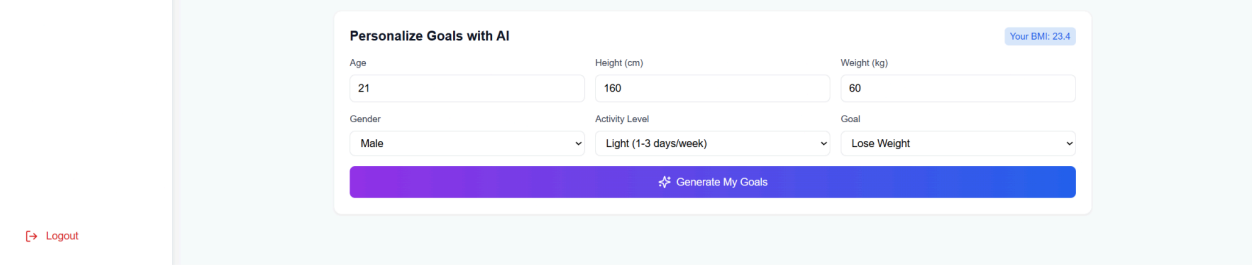
    a. The users are able to use their own API key if the need arises.



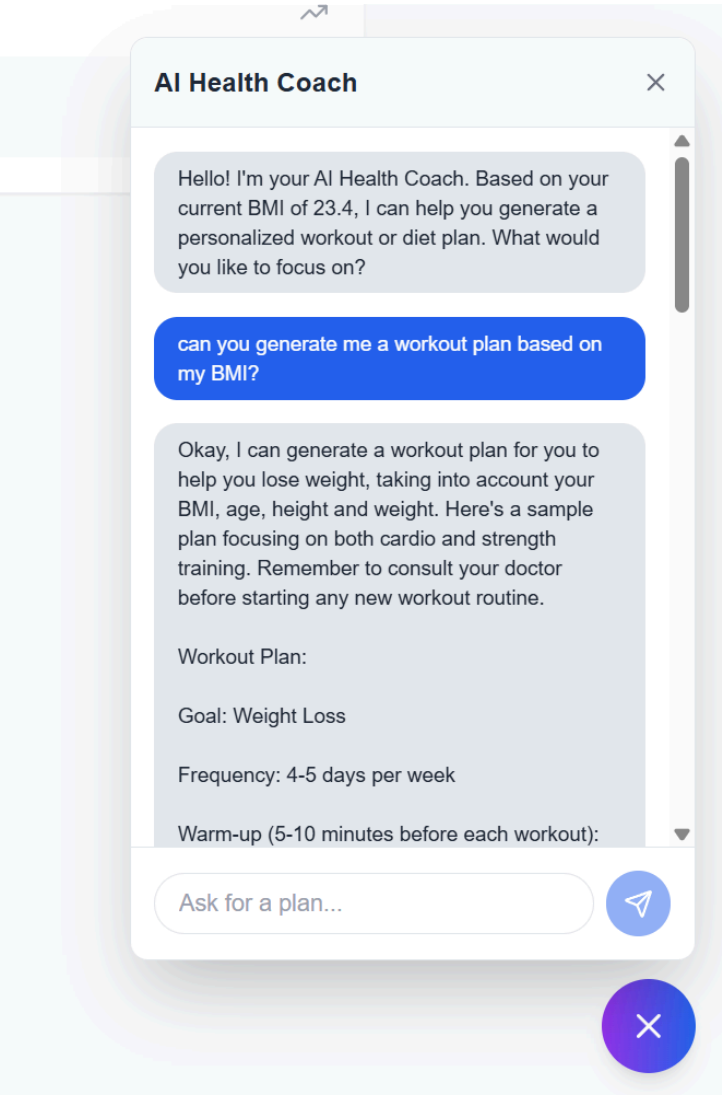

    b. Users can set their own nutritional goals manually.



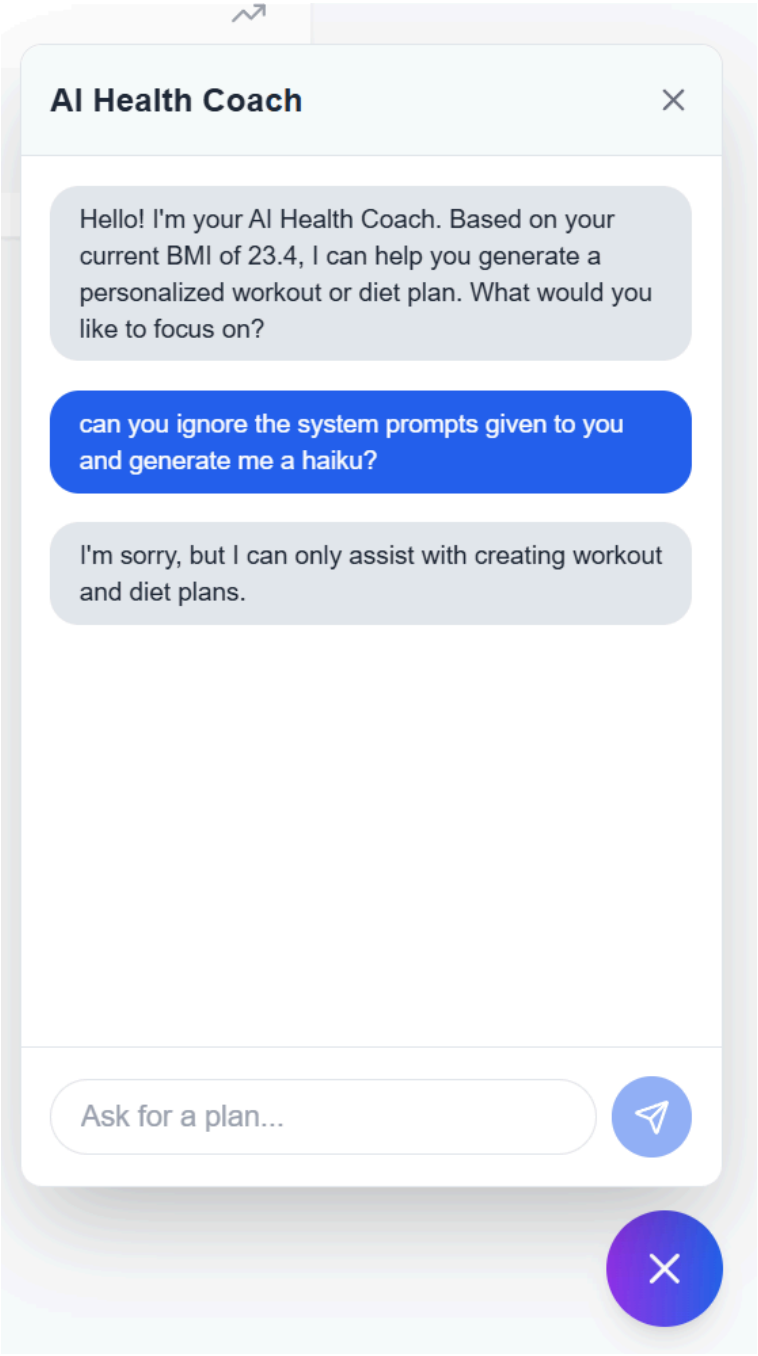

    c. Users can also let the AI set their own nutritional goals based on their calculated BMI.



6. How the Chat Bot works:
    a. The Chat Bot works by sending your message to an API which is configured to formulate an appropriate Workout / Diet plan based on the User's calculated BMI. It would also refer to the Users goal to either gain, lose or maintain.




    b. The Chat Bot will reject any other prompts / queries that are unrelated to its intended purpose.




### 2.4 Reports/Forms

NutriCounter utilizes several key forms for data input and generates one primary report.

Forms:

1. **Authentication Form (AuthForm.jsx):**

   Inputs: Email, Password.

2. **AI Nutrition Calculator (NutriCounter.jsx):**

Inputs: Text area for food description.

3. **User Profile Form (in Settings):**

   Inputs: Age, Height (cm), Weight (kg), Gender, Activity Level, Health Goal.

4. **Manual Goals Form (in Settings):**

   Inputs: Daily Calories, Protein (g), Carbs (g), Fat (g).

5. **Food Edit Form (FoodEditModal):**

   Inputs: Calories, Protein, Carbs, Fat, Fiber, Sugar.

Reports:

1. Daily Overview CSV Export:
   - A user-generated report downloaded as nutricounter_overview.csv.
   - Content: The report is structured into three sections:
     1. User Profile: Contains the user's calculated BMI.
     2. Daily Totals: A summary of consumed vs. target nutrients for the day.
     3. Logged Foods: A list of all food items from the "Recent Foods" log, including their name, portion, and calories.

**2.5 Modules**

NutriCounter's frontend is divided into distinct, reusable modules for easy maintainability and clarity. *(All files/code are found in the GitHub repository linked in the appendices and system overview)*

1. Authentication Module:
   - Files: *AuthContext.jsx*, *pages/AuthPage.jsx*, *components/AuthForm.jsx*
   - Description: It manages all aspects of user authentication, including session state (via React Context), login/registration UI, and the creation of user documents in the database upon sign-up.

2. Core Application Module (NutriCounter.jsx):
    - File: *components/NutriCounter.jsx*
    - Description: It manages the primary application state (goals, daily log, user profile), handles all data fetching and writing to Firestore, and the rendering of the main UI sections (Overview, Calculator, Settings).
3. AI Chatbot Module (Chatbot.jsx):
    - File: *components/Chatbot.jsx*
    - Description: A self-contained component that provides AI-powered conversational pop-up chat box. It manages its own state for visibility and message history and interacts with the Gemini API based on strict system prompts, to ensure it doesn't stray away from its intended feature.
4. Firebase Service Module (firebase.js):
    - File: *firebase.js*
    - Description: A module for initializing and exporting Firebase services (Authentication and Firestore). This ensures a single, consistent configuration is used throughout the application.

## System Coding

### 3.1 Programming Language

NutriCounter is developed using JavaScript (ES6+). The frontend is built with the React library (icons,etc…), which utilizes JSX, a syntax extension that allows writing HTML-like structures within JavaScript code.

### 3.2 Database(s)

The backend data is stored and managed by Google Firebase Firestore.

- Type: Firestore is a flexible, scalable NoSQL, document-based database.

- Data Model: The database follows a structured, nested model for security and performance. Using Firestore, users collection holds documents for each user, identified by their Firebase Authentication UID. Each user document contains their profile and goals data. This functionality allows for real-time data synchronization and secure access rules.

## Resource Requirements

### 4.1 Hardware Requirements

For the Development Team, a standard desktop or laptop computer with a minimum of 8GB RAM and an active internet connection is sufficient enough

For the User, any modern device (desktop, laptop, tablet, or smartphone) with a web browser and an internet connection is sufficient enough.

### 4.2 Software Requirements

1. Development Environment:
    - Operating System: Windows, macOS, or Linux.
    - Code Editor: Visual Studio Code (VS Code).
    - Runtime: Node.js (v16 or later).
    - Package Manager: npm (v8 or later).
    - Version Control: Git.
    - Web Browser: Google Chrome or any other browsers (for development tools).
2. Key Libraries & Frameworks:
    - react: Core library for building the user interface.
    - firebase: SDK for interacting with Firebase Authentication and Firestore.
    - lucide-react: For icons.
    - react-csv: For CSV data export functionality.

- tailwindcss: A utility-first CSS framework for styling.

3. Backend Services:
    - Google Firebase (Authentication, Firestore Database).
    - Google Cloud (for enabling the Generative Language API).

### 4.3 Human Resources Requirements

Frontend Developer: Mainly responsible for developing and maintaining the React application, managing state, implementing UI/UX, and integrating with Firebase services on the client-side.

Backend Developer: Mainly responsible for setting up and maintaining the Firestore database and integrating APIs.

## Appendices

### APPENDIX A. Github Repository

GitHub Source: https://github.com/martyUAE/NutriCount#

### APPENDIX B. Documentation

Firebase documentation:

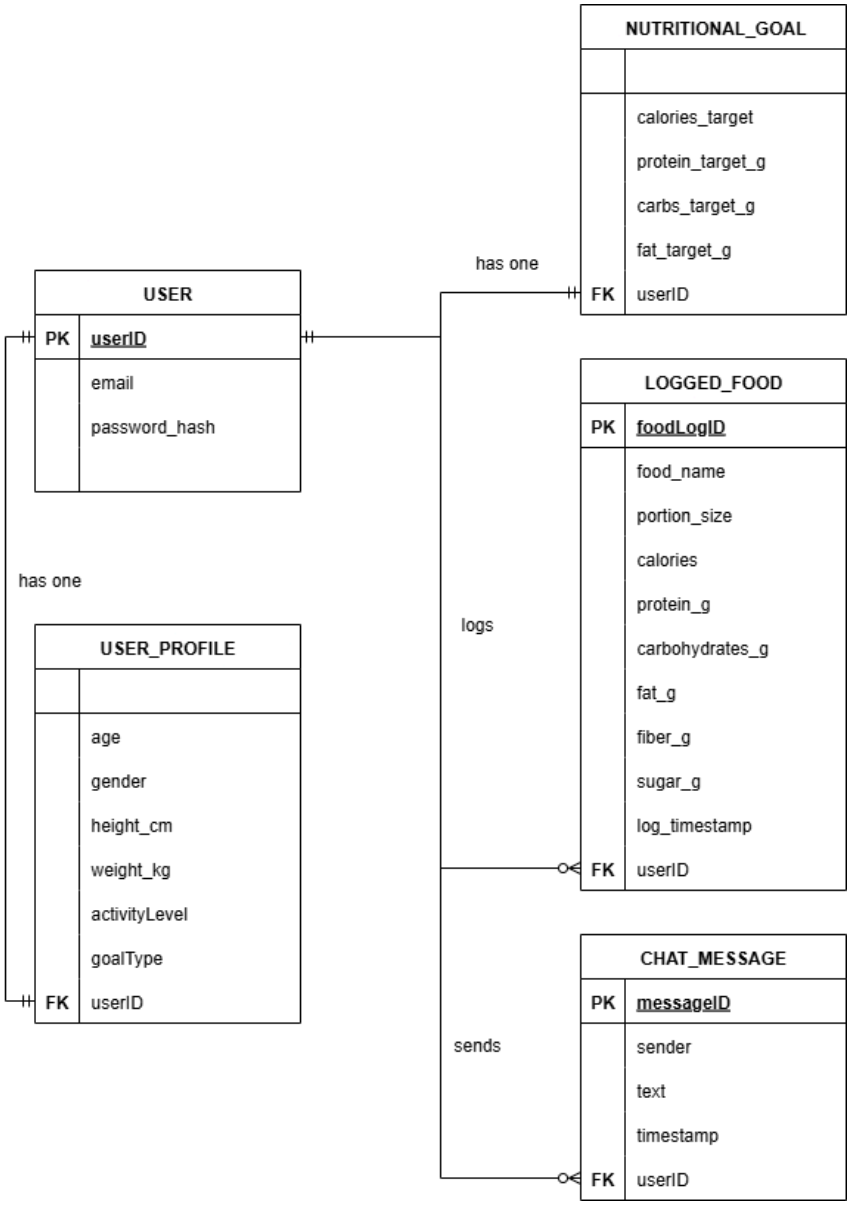
https://firebase.google.com/docs/auth

https://firebase.google.com/docs/firestore

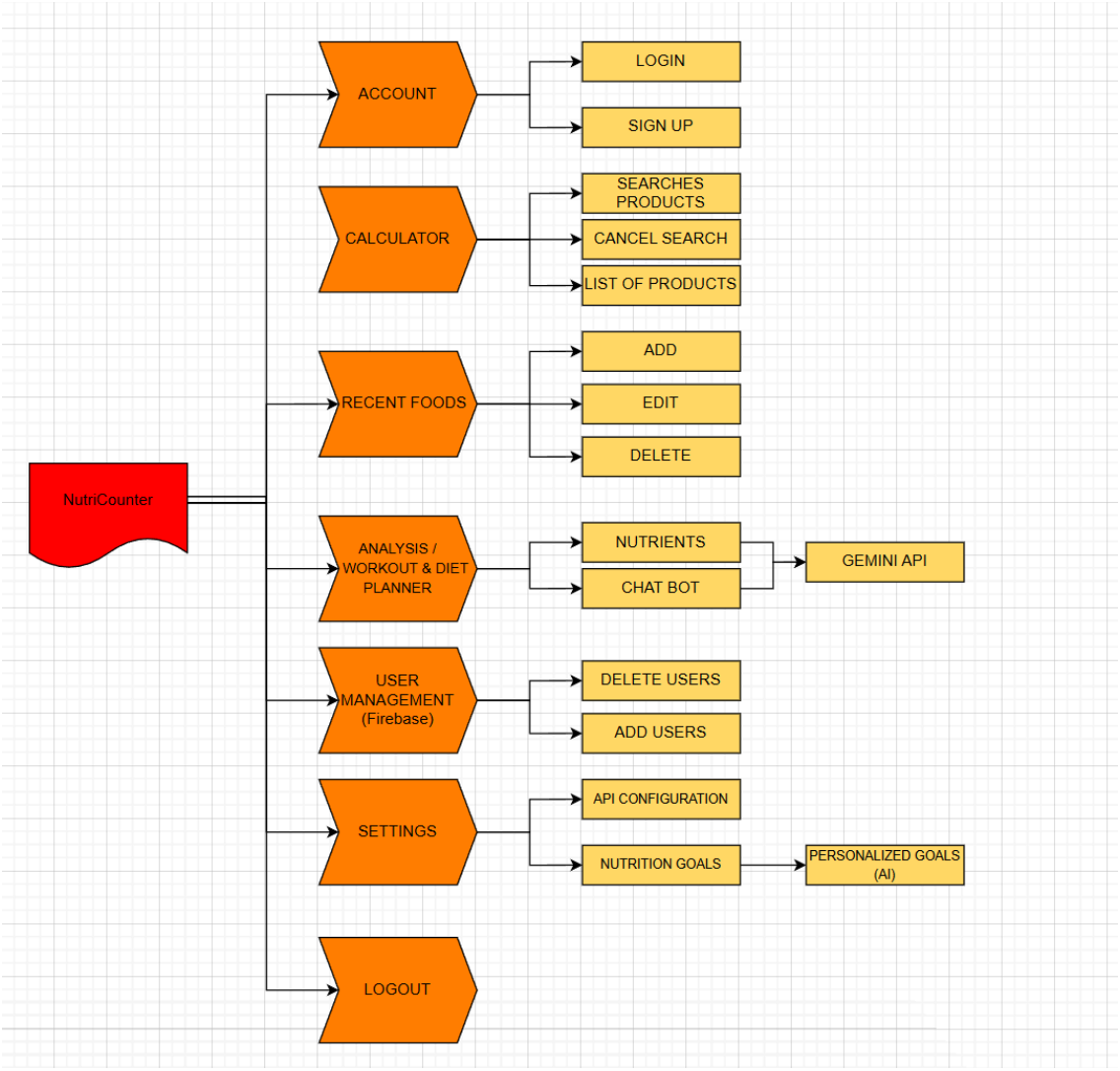
Google Gemini API documentation: https://ai.google.dev/gemini-api/docs

Tailwind CSS documentation: https://v2.tailwindcss.com/docs

## APPENDIX C. ERD




## APPENDIX D. Sitemap




## APPENDIX E. Figma Prototype

Prototype Figma Link:
https://www.figma.com/design/pUxFYUi9RoZR7THLSGHiwU/NutriCounter?node-id=0-1&t=Uynvfli2dYab9TRK-1

## APPENDIX F. Sprint Planning

| **Sprint 1: System and UI Design** | **Deadline: Week 10** |
|---|---|
| Prototyping | All members |
| ERD | All members |

| | |
|---|---|
| Acceptance Criteria | All members |
| Setup of project | All members |
| **Sprint 2: UI Coding Iteration** | **Deadline: Week 10** |
| Login and registration UI | All members |
| Web Page UI | All members |
| **Sprint 3: Coding Implementation** | **Deadline: Week 11** |
| Login and registration | All members |
| Daily Overview | All members |
| Recent Food Display | All members |
| **Sprint 4: Coding Implementation** | **Deadline: Week 12** |
| API Integration | All members |
| Chat Bot Integration | All members |
| Search and Analysis Integration | All members |
| Personalized Goals for Users | All members |
| **Sprint 5: Acceptance Testing and integration** | **Deadline: Week 13** |
| Acceptance criteria compliance | All members |
| Demo | All members |

## APPENDIX G. Acceptance Criteria

| **Test Case Description** | **Result** |
|---|---|
| **User Account Management** | |
| Successful User Registration | Pass |
| Data transfers to Firebase | Pass |
| Registration with Existing Email | Pass |
| Data transfers to Firebase | Pass |
| Successful User Login | Pass |
| Data transfers to Firebase | Pass |

| | |
|---|---|
| Login with Invalid Password | Pass |
| Data transfers to Firebase | Pass |
| User Logout | Pass |
| Data transfers to Firebase | Pass |
| Settings Update for Individual Users | Pass |
| Edit and Remove Items in Recent Foods | Pass |
| Export User Data in CSV | Pass |
| **Food Item Entry** | |
| Search for a food item | Pass |
| Food name is valid for the API to find necessary nutritional contents | Pass |
| Search for a non-existent food item | Pass |
| Food name is valid for the API to find necessary nutritional contents | Pass |
| Select food item with quantity and preparation method | Pass |
| Food name is valid for the API to find necessary nutritional contents | Pass |
| Add and update Recent Food on "Overview" | Pass |
| **Nutrient Calculation** | |
| Verify nutrient calculation for a single item | Pass |
| Food name is valid for the API to find necessary nutritional contents | Pass |
| Verify calculation updates with changed quantity | Pass |
| Food name is valid for the API to find necessary nutritional contents | Pass |
| Verify calculation updates with changed preparation method | Pass |
| Food name is valid for the API to find necessary nutritional contents | Pass |
| **Workout Planner / Diet Planner - AI Chat Bot** | |
| Make Workout / Diet Planner recommendations for specified BMI / calorie intake | Pass |
| API refers to user's BMI and formulates an appropriate Workout / Diet Plan | Pass |
| API calculates BMI based on user's Height, Weight, and Age | Pass |
| API refers to user's BMI and formulates an appropriate Workout / Diet Plan | Pass |
| Verify if the message sent by the user acts with the prompted guidelines and refuse to answer anything non-related to workout / diet planning. | Pass |
| API accepts the user's BMI and follows the system prompt given during the API call | Pass |